

Assignment:

Load any dataset and Explore sampling techniques

We will use students.csv dataset

Simple Random Sampling

```
sample_random = df.sample(n=10)

print(sample_random)
```

Systematic Sampling

(every 5th row)

```
systematic_sample = df.iloc[::5]

print(systematic_sample)
```

Stratified Sampling

```
from sklearn.model_selection import train_test_split

sample_data, remaining = train_test_split(
    df,
    test_size=0.8,
    stratify=df['Gender'],
    random_state=1
)

print(sample_data['Gender'].value_counts())
```

Cluster Sampling

```
cluster_sample = df[df['City'] == 'Delhi']

print(cluster_sample)
```